10B — Deep-Dive Audit: Economy & Supply Chain (Part 2 of 5: Workshop Queues, 4-Discipline Crafting, `craftTransactions` Refund Leak, `EC-10` Double Salvage, Recipe Lifecycle & Craft XP)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'`)
- **Primary Audited Modules**: `js/economy.js`, `js/ai.js`, `js/sect.js`, `js/sim.js`, `js/content.js`
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini**)
- **Baseline Filter**: Excludes `EC-01..04, EC-07` already fixed in `v2.4.0`; addresses confirmed live `EC-10` plus 5 newly uncovered structural crafting/queue defects in `JS2-v2.6.0`.

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1. `[ECO-WORK-01]` Single-Slot `orders.forge` Starves `TALISMAN` & `ARRAY` Production (`js/economy.js:420–480`, `js/sect.js`)

1.1 Three Crafts Competing for One Order Variable (`W.sect.orders.forge`)

- **Code Trace**:
  - `SECT.hallHeads` assigns `W.sect.orders.alchemy` (`['ALCHEMY']`) and `W.sect.orders.forge` (`['FORGING', 'ARRAY', 'TALISMAN']`).
  - Inside `ECO.pickOrder`, weapon and armor shortages receive large priority weights (`noWeapon * 1.4`, `lowTierWeapon * 0.9`), whereas Talisman (`REC190_TAL_*`) and Formation Core (`REC190_CORE_*`) recipes receive **`0` demand weight** and suffer the tier penalty `-(r.tier - 3) * 0.12`.
- **Second-Order Idle-Specialist Cascade**:
  - Whenever `W.sect.orders.forge` holds a Blacksmith weapon recipe (`craft: 'FORGING'`), every dedicated **`ARRAY_TALISMAN`** / **`INK_LITERATI`** / **`OCCULT_MAGE`** **disciple (`TALISMAN = 85, ARRAY = 80, FORGING = 4`)** fails `ECO.craftEligible` on `orders.forge.rid`!

    - Even though `forge` physically contains dedicated `'array'` benches (`world_map.js:97`) and `W.sect.stock` is full of Cinnabar (`MAT_ORE_CINNABAR_1`) and Jade Slates, Talisman and Array specialists are completely locked out of sect crafting!
- **Remediation Blueprint**:
    - Split `W.sect.orders` into **three independent hall queues**: `orders.alchemy` (`['ALCHEMY']`), `orders.forge` (`['FORGING']`), and **`orders.array_talisman` (`['ARRAY', 'TALISMAN']`)**.
    - Add real situational demand into `ECO.needSummary` for `ARRAY` & `TALISMAN`: damaged `elementalBarrier` (`< 75%` HP) boosts `REC190_CORE_009`, bottlenecked major-stage disciples boost `REC190_TAL_019` / `CORE_008` (`Tribulation Shield`), and active frontier wars/expeditions boost combat & escape talismans.

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2. `[ECO-WORK-02] 475` Registered Expansion Recipes Invisible to `ECO.recipesFor` & High-Tier Penalty Inversion (`js/economy.js:88, 454`, `js/sim.js:545`)

1. **`475` Expansion Recipes Skipped by `ECO.recipesFor` (`economy.js:88`)**:
    - `ECO.recipeById` contains `1,370` recipes, but `ECO.recipesFor(W, crafts)` iterates `JD.ITEMS.recipes` (`895` base entries), omitting **`475` expansion/Grand-Dao/ecology recipes** from autonomous hall-head production (`0%` autonomous orders). Iterate a deduplicated `ECO.allRecipes` array synchronized with `ECO.recipeById`.
2. **High-Tier Recipe Penalty (`s -= (r.tier - 3) * 0.12`, `economy.js:454`)**:
    - Once a sect unlocks `recipeTier = 6`, `pickOrder` penalizes Tier-6 recipes by `-0.36` compared to Tier-3 recipes, causing wealthy endgame workshops to spam Tier-1..2 gear instead of upgrading elders to Tier-5..6 equipment! Invert this when `need.undergearedElders > 0` and inputs above `reserveFloor` are available.
3. **`SIM.researchTick` Ignores `FORGING`, `ARRAY`, & `TALISMAN` Skills (`sim.js:545`)**:
    - `(E(head, 'intellect.comprehension') + sk.SCRIPTURE + sk.ALCHEMY) / 10` omits `Math.max(sk.ALCHEMY || 0, sk.FORGING || 0, sk.ARRAY || 0, sk.TALISMAN || 0)`, penalizing non-alchemist Scripture Pavilion Heads.

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3. [ECO-CRAFT-03] Aborted-Craft Input Destruction on Raid Interrupts & Unfixed EC-10 Double Salvage (js/economy.js & js/ai.js)

3.1 Raid/Emergency Interrupts Permanently Destroy Reserved Tier-5..7 Crafting Inputs!

- **Code Trace (`ECO.reserveInputs` $\to$ `AI.interruptFor` / `AI.cancelAct` $\to$ `ECO.pruneCraftTransactions`)**:
  - At `brew.start` / `forge.start`, `ECO.reserveInputs` immediately deducts all recipe materials from `W.sect.stock` and creates `craftReceipt = { state: 'reserved', inputs }`.
  - If a **Sect Raid (`COMBAT.enter` $\to$ `AI.interruptFor`)**, **Injury**, or **Starvation** cancels `n.act` mid-craft, `AI.cancelAct` clears `n.act` **without refunding `craftReceipt.inputs`**! Later, `ECO.pruneCraftTransactions` deletes the stale `'reserved'` entry, **permanently vaporizing scarce Tier-5..7 herbs, ores, and beast cores**!
- **Remediation Blueprint**:
  - In `AI.cancelAct(W, n)` (and `ECO.abortCraftReceipt(W, receiptId)`): whenever an active `'brew'`, `'forge'`, `'personalBrew'`, or `'personalForge'` action with `ctx.craftReceipt?.state === 'reserved'` is interrupted before completion:
    - If interrupted before `40%` progress (or interrupted by external combat call-to-arms), **refund `100%` of `entry.inputs`** back to the originating source (`W.sect.stock` or `n.inv.mats`) and mark `entry.state = 'aborted_refunded'`.
    - If interrupted past `40%` due to personal neglect, refund primary non-perishable ores (`100%`) and `50%` of herbs.

3.2 Unfixed EC-10 Double-Salvage on Stale Object References (economy.js:1932–1973)

- **Live Defect**: Calling `ECO.retireGear(W, null, gearObj)` twice with the same object reference yields `+2` salvage ores (`MAT_ORE_BLACK_IRON_1` / `COLD_STEEL_2`) even though the item was already removed from `W.sect.gear` on the first call!
- **Exact Fix**: Require locating `gearObj` by `uid`/reference inside `npc.equip`, `npc.inv.gear`, or `W.sect.gear` (`loc !== null`), reject if `gearObj._salvaged === true`, set `gearObj.retired = true; gearObj._salvaged = true;` atomically before adding salvage material, and return material matching `gearObj.origin?.materials` (`wood/silk/jade/iron`, `05F`)!

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4. [ECO-PERS-04] ALCHEMY-Only Lockout in ECO.personalCraft (js/economy.js:297) & Recipe Scroll Social Flow (js/content.js)

1. **Blacksmiths, Scribes & Arrayists Locked Out of Personal Crafting (economy.js:297)**:
   - `if (r.outType !== 'PILL') return null;` blocks `FORGING`, `TALISMAN`, and `ARRAY` disciples from ever crafting personal equipment, talismans, or tools using their own `n.inv.mats` or personal CP!
   - **Fix**: Introduce `A.personalForge` (mirroring `A.personalBrew`) allowing `FORGING`, `TALISMAN`, and `ARRAY` practitioners to craft personal upgrades (`'useful-now'`) or apprentice/companion gifts (`'gift-worthy'`) from `n.inv.mats`!
2. **Physical Proximity for Peer Recipe Barter (content.js)**:
   - Require two NPCs to be co-located (`dist <= 4` in `tea_garden` / `market` / `plaza`) rather than telepathically scanning `b.inv.stones` across the map (`02A`), and let loyal discoverers (`loyalty >= 62`) transcribe `n.inv.recipes` into `W.sect.knownRecipes` at `scripture` for `CPE.reward` + Chronicle credit!

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5. [ECO-CULT-05] Craft Cultivation (ECO.grantCraftCultivation) Yields 10×–18× Less XP Per Hour Than Idle Cushion Meditation (js/economy.js:377–395)

- **Math Proof**:
  - `ECO.craftHours` for a Tier-4 item takes `~9.5` hours (`570` minutes), granting `basePts = (2.2 + 4 * 1.65) * 1.1 = 9.7` cultivation points (`0.017 XP/min` on `CRAFT_CREATION`, and `0.004 XP/min` on `ELIXIR_DAO`!).
  - Meditating for `570` minutes yields `~120–210` XP (`0.21–0.37 XP/min` — **`12×` to `50×` faster**!). Thus, dedicated Alchemists and Blacksmiths fall 2–3 full realms behind peers simply by doing their jobs!
- **Remediation Blueprint**:
  - Anchor `grantCraftCultivation` to **time invested × craft tier × quality**: `pts = craftMinutes * CULT.rate(npc) * pathCraftMult * (0.85 + 0.12 * r.tier) * clamp(q, 0.8, 1.5)`, where `pathCraftMult` is **`1.15×`** for `CRAFT_CREATION` (making forging/brewing their *primary* Dao

cultivation!), **`0.85×`** for `ELIXIR_DAO` / `RUNE_ARRAY_ENGRAVER` / `SPIRIT_HUSBANDRY`, and **`0.35×`** for general paths!